
USV Events Installation Guide

1. Prerequisites

Docker Desktop 4.x or newer

Git

Optional for local development without Docker:

- Python 3.12+ - Node.js 20+ - npm 10+

2. Quick Start With Docker

1. Clone or open the repository.

2. Create a root `.env` file if you want to override the default values from `docker-compose.yml`.

3. Start the stack:

```
docker compose up --build
```

4. Open the services:

Frontend: <http://localhost:5173>

Backend API: <http://localhost:8000>

Swagger UI: <http://localhost:8000/api/v1/docs>

PostgreSQL: <localhost:5432>

3. Environment Variables

The backend reads configuration from environment variables. The most important ones are:

`DATABASE_URL`

`SECRET_KEY`

`APP_ENV`

`FORCE_HTTPS_REDIRECT`

`ALLOWED_HOSTS`

`GOOGLE_CLIENT_ID`

`GOOGLE_CLIENT_SECRET`

`GOOGLE_REDIRECT_URI`

`FRONTEND_URL`

`FRONTEND_RESET_PASSWORD_PATH`

`SMTP_SERVER`

`SMTP_PORT`

`SMTP_USER`

```
SMTP_PASSWORD
SMTP_FROM_EMAIL
SMTP_FROM_NAME
ENABLE_EMAIL
```

A ready-to-use example is available in `backend/.env.example`.

4. Database Initialization

The backend can run database setup automatically on startup:

```
migrations for registration status
idempotent demo seed data
```

You can control this behavior with:

```
AUTO_MIGRATE_ON_STARTUP=true
AUTO_SEED_ON_STARTUP=true
```

If you already have a running stack and need to apply the registration status migration once, run:

```
docker exec usv_events_backend python -m app.migrate_event_registration_status
```

5. Email Configuration

Email notifications are disabled by default.

To enable them:

1. Configure your SMTP provider.
2. Set `ENABLE_EMAIL=true`.
3. Provide `SMTP_SERVER`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, and sender details.

Common providers work with the same settings model:

```
Gmail SMTP
Microsoft 365 / Outlook
SendGrid SMTP
Mailgun SMTP
```

For provider examples and troubleshooting, see `backend/docs/EMAIL_CONFIGURATION.md`.

6. Production Hardening

Before deploying to a production environment:

1. Set `APP_ENV=production`.
2. Replace the default `SECRET_KEY` with a long random value.

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3. Keep `FORCE_HTTPS_REDIRECT=true` if you want the backend to enforce HTTPS.
 4. Restrict `ALLOWED_HOSTS` to your real domain names.

When `APP_ENV=production`, the backend fails fast if it still uses the placeholder secret key.

7. Google OAuth Setup

If you want Google login for students:

1. Create OAuth credentials in Google Cloud Console.
2. Set the authorized redirect URI to:

```
http://localhost:5173/auth/callback
```

3. Fill in `GOOGLE_CLIENT_ID` and `GOOGLE_CLIENT_SECRET` in your environment.

If the Google settings are missing, the Google login flow will return a configuration error.

8. Local Development Without Docker

Backend

```
cd backend
python -m venv .venv
```

Activate the environment:

```
# Windows
.venv\Scripts\activate

# Linux/macOS
source .venv/bin/activate
```

Install dependencies and start the API:

```
pip install -r requirements.txt
uvicorn main:app --reload --port 8000
```

Frontend

```
cd frontend
npm install
npm run dev
```

9. PDF Guide Generation

This repository includes a small generator that builds the PDF version of this guide from the markdown source.

Run it from the repository root:

```
npm run docs:pdf
```

The generated file is written to:

```
docs/installation-guide.pdf
```

10. Stopping The Stack

```
docker compose down
```

To remove volumes and reset the database too:

```
docker compose down -v
```

11. Troubleshooting

If port 5432 is already in use, stop the local PostgreSQL service or change the mapped port in `docker-compose.yml`.

If login fails, verify `SECRET_KEY`, Google OAuth settings, and the database connection string.

If email sending fails, keep `ENABLE_EMAIL=false` in development until the SMTP settings are valid.

If Docker rebuilds slowly, make sure the volumes are healthy and the backend image has been rebuilt after dependency changes.

12. Recommended First Run Checklist

- ☐ Copy or create `.env`
- ☐ Confirm Docker Desktop is running
- ☐ Start the stack with `docker compose up --build`
- ☐ Open the frontend in the browser
- ☐ Verify the backend health endpoint
- ☐ Configure Google OAuth if student login is needed
- ☐ Configure SMTP if email notifications are needed